

NEHBANDAN, Iran, Islamic Republic of

WMO: 408270

Lat: 31.5416N

Lon: 60.0377E

Elev: 3973

StdP: 12.71

Time Zone: 3.50 (IRN)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.6	(c) 27.8	(d) -15.1	(e) 2.8	(f) 66.9	(g) -8.1	(h) 4.1	(i) 64.9	(j) 18.3	(k) 52.5	(l) 16.2	(m) 50.6	(n) 0.9	(o) 90	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.8	(c) 105.4	(d) 61.8	(e) 103.3	(f) 61.5	(g) 101.2	(h) 60.6	(i) 66.1	(j) 97.9	(k) 64.4	(l) 97.7	(m) 63.0	(n) 95.9	(o) 9.1	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.5	70.5	79.0	50.9	64.0	72.2	48.6	58.8	71.3	33.1	97.8	31.6	97.5	30.5	96.4	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 27.1	(b) 22.9	(c) 19.5	(e) 19.7	(f) 109.4	(g) 3.8	(h) 1.3	(i) 17.0	(j) 110.3	(k) 14.8	(l) 111.0	(m) 12.7	(n) 111.8	(o) 10.0	(p) 112.7		
			DB													
			WB													
				16.4	66.6	3.2	3.2	14.0	68.9	12.2	70.8	10.3	72.6	8.0	74.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.0	44.1	49.5	60.5	71.2	81.1	88.8	91.4	87.5	80.3	69.9	56.2	46.6
	DBStd	17.46	6.68	7.79	7.70	6.14	5.06	3.79	3.76	3.96	4.60	5.99	6.89	5.77
	HDD50	460	203	91	11	0	0	0	0	0	0	0	20	135
	HDD65	2140	648	433	178	17	0	0	0	0	0	21	273	570
	CDD50	7410	20	79	337	637	965	1165	1282	1161	910	617	207	30
	CDD65	3615	0	1	39	204	500	715	817	696	460	173	10	0
	CDH74	50183	0	19	462	2229	6530	10762	12588	9699	5618	2112	162	2
CDH80	29322	0	0	116	854	3639	6813	8276	5732	3018	850	24	0	
Wind	WSAvg	7.4	4.2	5.2	6.0	5.9	7.8	10.9	13.9	13.0	8.1	4.9	4.0	4.2
Precipitation	PrecAvg	4.8	1.1	1.1	1.4	0.5	0.1	0.0	0.0	0.0	0.0	0.1	0.1	0.5
	PrecMax	11.5	2.4	3.7	5.1	2.0	0.7	0.2	0.1	0.0	0.2	0.7	0.9	3.0
	PrecMin	0.6	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.5	0.6	1.1	1.5	0.6	0.2	0.0	0.0	0.0	0.0	0.2	0.2	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.9	77.5	89.5	94.6	103.3	107.8	107.9	105.6	100.4	95.6	83.4	72.1
		MCWB	48.2	51.0	53.0	56.6	59.8	62.4	62.7	62.2	57.3	58.4	54.9	48.7
	2%	DB	64.3	71.7	82.6	91.5	100.1	105.2	105.7	102.2	98.0	91.8	78.5	66.8
		MCWB	45.3	49.8	52.7	56.7	60.9	62.2	61.5	61.2	57.6	58.1	53.0	46.2
	5%	DB	61.5	67.4	78.9	88.2	97.3	102.9	103.8	99.7	95.6	88.2	74.6	63.8
		MCWB	44.1	48.3	52.0	56.3	60.7	61.7	61.6	60.5	57.4	56.4	51.6	44.8
	10%	DB	58.0	63.8	75.3	85.1	94.6	100.5	101.7	97.3	93.3	84.9	70.7	60.3
		MCWB	42.9	46.8	51.3	55.2	59.7	60.4	60.7	59.4	57.1	55.2	49.8	42.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	54.4	57.6	60.4	68.9	67.2	67.5	68.3	61.6	61.0	57.5	50.3
		MCDB	60.0	67.5	74.5	86.0	94.0	99.3	102.7	95.7	94.2	89.2	76.0	67.0
	2%	WB	48.7	52.2	55.9	58.7	64.1	65.6	65.4	64.6	60.2	59.2	55.2	48.0
		MCDB	60.5	65.4	73.1	84.4	92.6	97.8	100.3	96.9	92.7	88.6	73.4	64.0
	5%	WB	46.6	50.6	54.3	57.6	62.1	64.1	63.8	62.7	59.1	57.7	53.2	46.1
		MCDB	57.7	63.3	71.9	82.6	92.7	97.1	99.1	95.7	91.3	85.9	70.5	60.4
	10%	WB	44.2	48.6	52.8	56.5	60.6	62.6	62.4	60.9	57.9	56.2	51.6	44.1
		MCDB	55.0	61.1	70.7	80.4	90.8	95.2	96.8	94.0	89.7	81.8	67.4	57.8
Mean Daily Temperature Range	5% DB	MDBR	22.6	22.2	23.3	23.2	24.2	22.2	19.8	20.0	25.9	27.4	25.4	25.8
		MCDBR	26.8	25.8	27.4	25.9	27.0	25.3	21.5	21.1	27.8	29.3	28.6	29.0
		MCWBR	14.5	11.8	10.7	8.9	9.2	8.0	6.9	6.8	10.2	11.8	13.1	15.1
	5% WB	MCDBR	22.1	20.0	21.5	22.9	24.2	23.3	21.6	19.8	26.1	27.6	24.7	25.7
		MCWBR	12.7	9.9	8.8	7.8	8.7	8.1	7.7	7.3	10.3	11.3	11.7	14.2
Clear-Sky Solar Irradiance	taub		0.324	0.373	0.435	0.496	0.509	0.474	0.445	0.412	0.383	0.391	0.351	0.315
	taud		2.341	2.165	1.970	1.846	1.832	1.918	2.008	2.100	2.160	2.122	2.259	2.379
	Ebn at Noon		284	278	267	255	252	260	267	275	280	267	270	281
	Edh at Noon		32	43	56	66	67	61	56	50	45	44	35	30
All-Sky Solar Radiation	RadAvg		1130	1373	1742	2063	2334	2595	2547	2413	2100	1660	1257	1077
	RadStd		83	94	99	92	97	75	51	61	44	51	57	69

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.76	N/A	-6.30	+1.28	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page